

# Continuous Bag-of-Words (CBOW)

Dr Partha Majumdar

Department of Computer Science,  
Swiss School of Business and Management  
(SSBM), Geneva, Switzerland

ORCID: 0009-0002-7375-8034



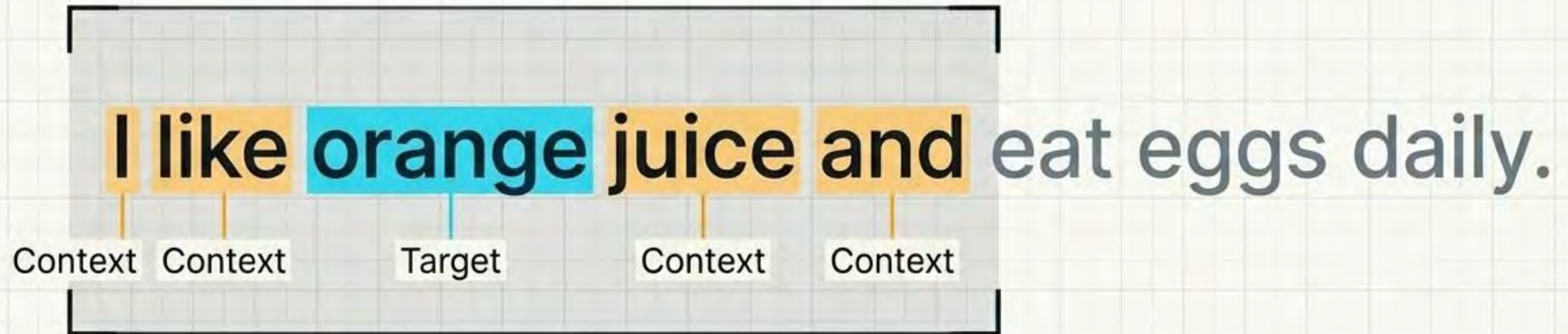
# The Evolution of Language Representation

| Paradigm         | Unit Treatment               | Vector Type            | Geometric Distance                             |
|------------------|------------------------------|------------------------|------------------------------------------------|
| One-Hot Encoding | Independent Entities         | Extreme Sparsity       | None<br>(No semantic awareness)                |
| TF-IDF           | Frequency & Rarity Based     | Statistically Weighted | Statistical<br>(Word importance, not context)  |
| CBOW             | Context-Driven Co-occurrence | Dense Embeddings       | Semantic<br>(Meaning equals spatial proximity) |

Meaning is no longer assigned or computed directly; it is learned implicitly through patterns of co-occurrence.



# Framing Language as a Prediction Task

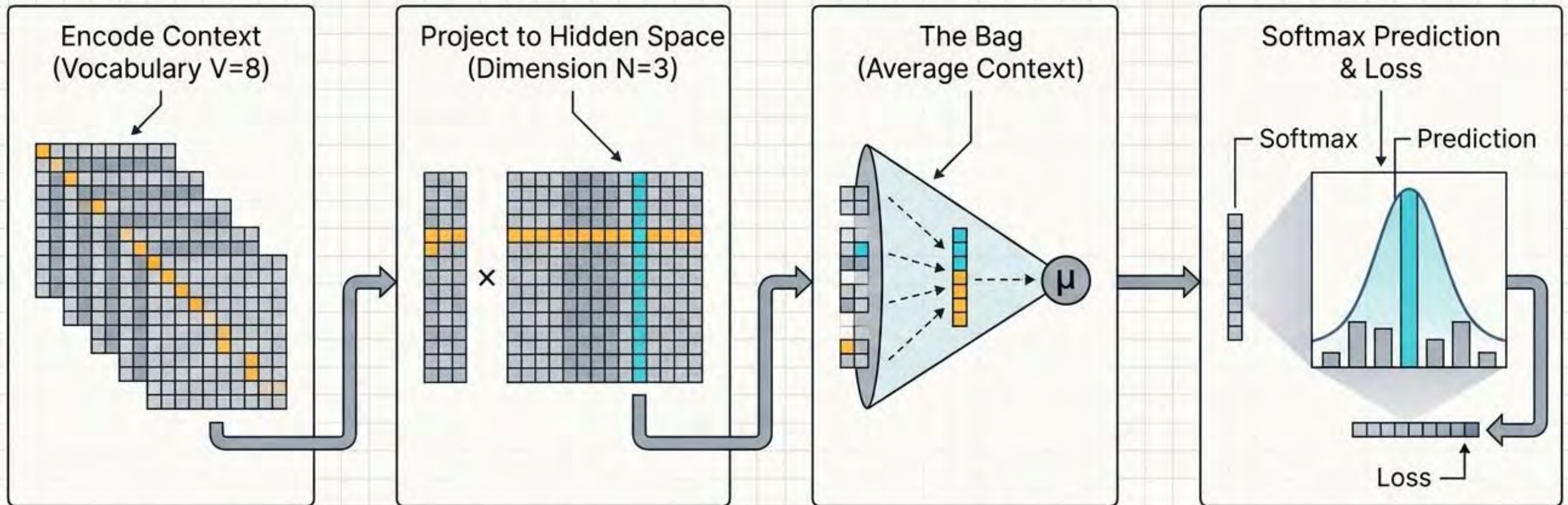


Symmetric window of size 2. The surrounding words become the input, and the central word becomes the expected output.

**Takeaway:** The model is not explicitly told what juice means. It extracts meaning from its company.



# The Forward Pass Pipeline



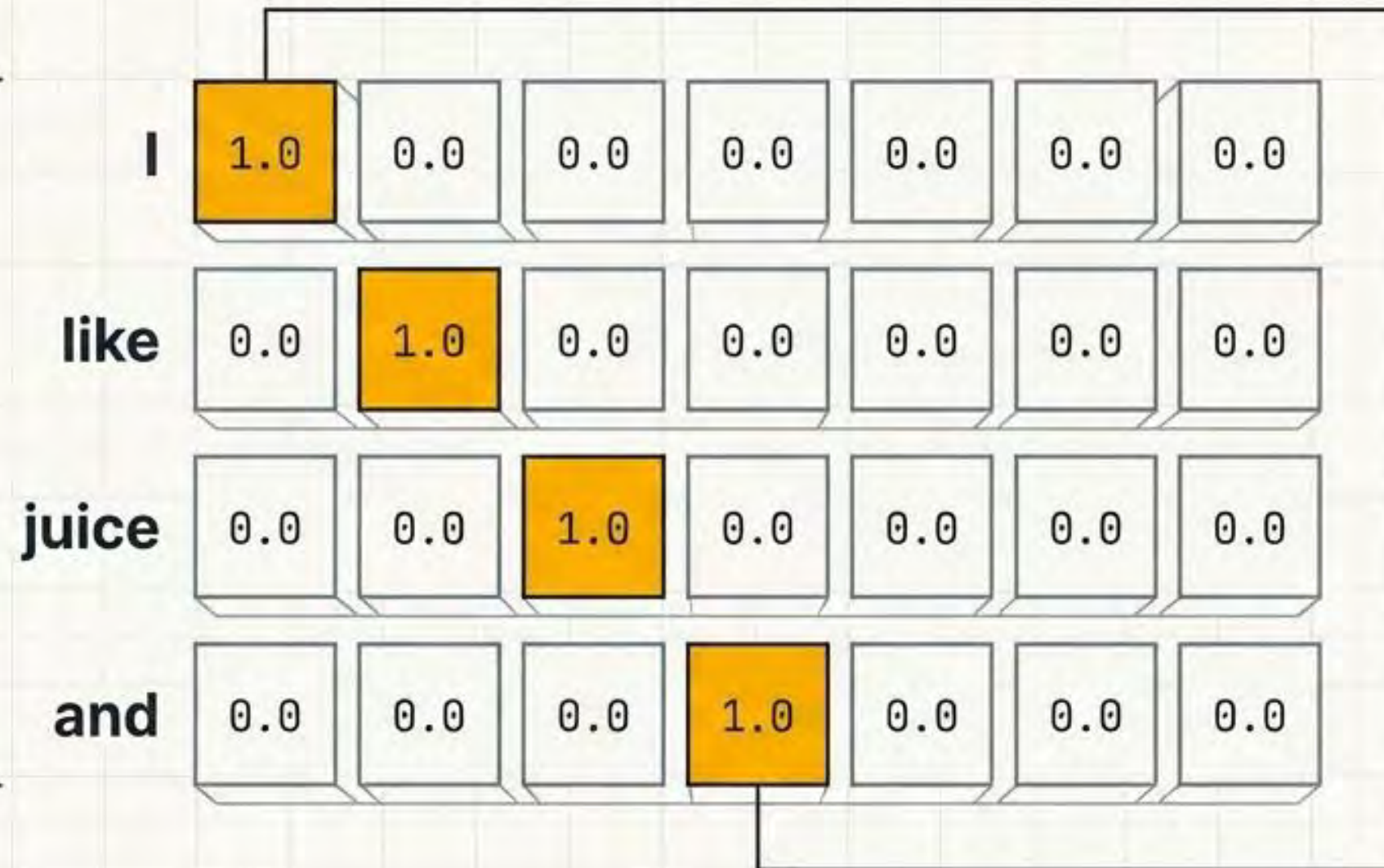
sentence = I like orange juice and eat eggs daily



# Step 1: Input Encoding

Vocabulary dictionary  
assigns an index to each  
of the  $V=8$  words.

['I', 'like', 'orange',  
'juice', 'and', 'eat',  
'eggs', 'daily']



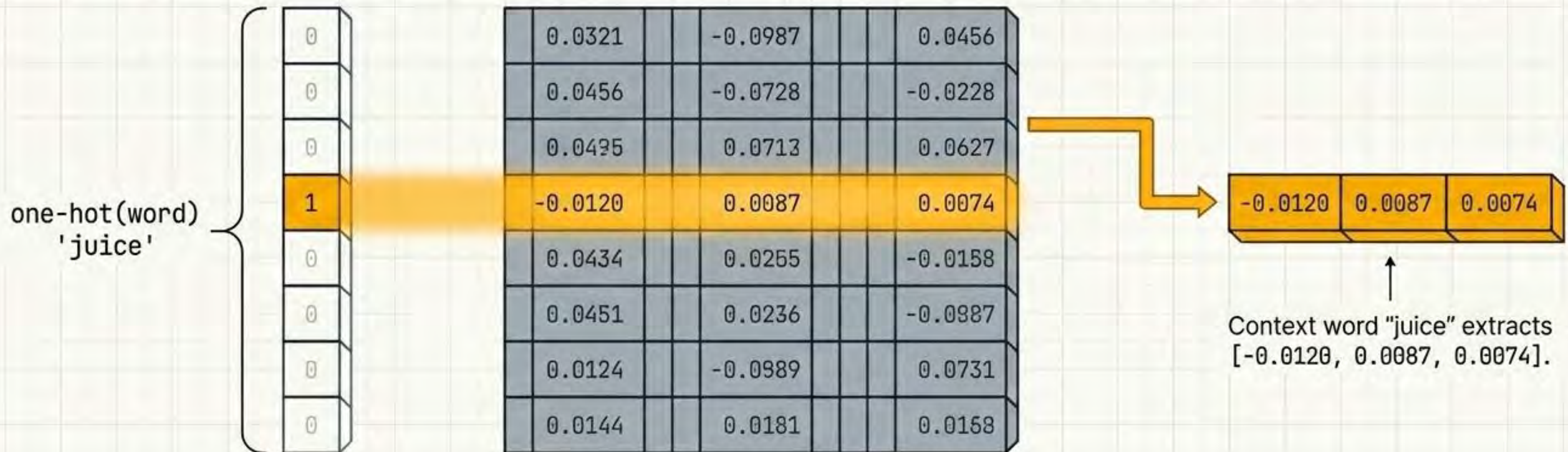
One-Hot representation  
ensures each word is  
structurally independent  
before entering the  
network.



# Step 2: The Lookup Mechanism

$$\text{one-hot}(\text{word}) @ W = \text{embedding}(\text{word})$$

## Matrix Multiplication as a Lookup Tool.

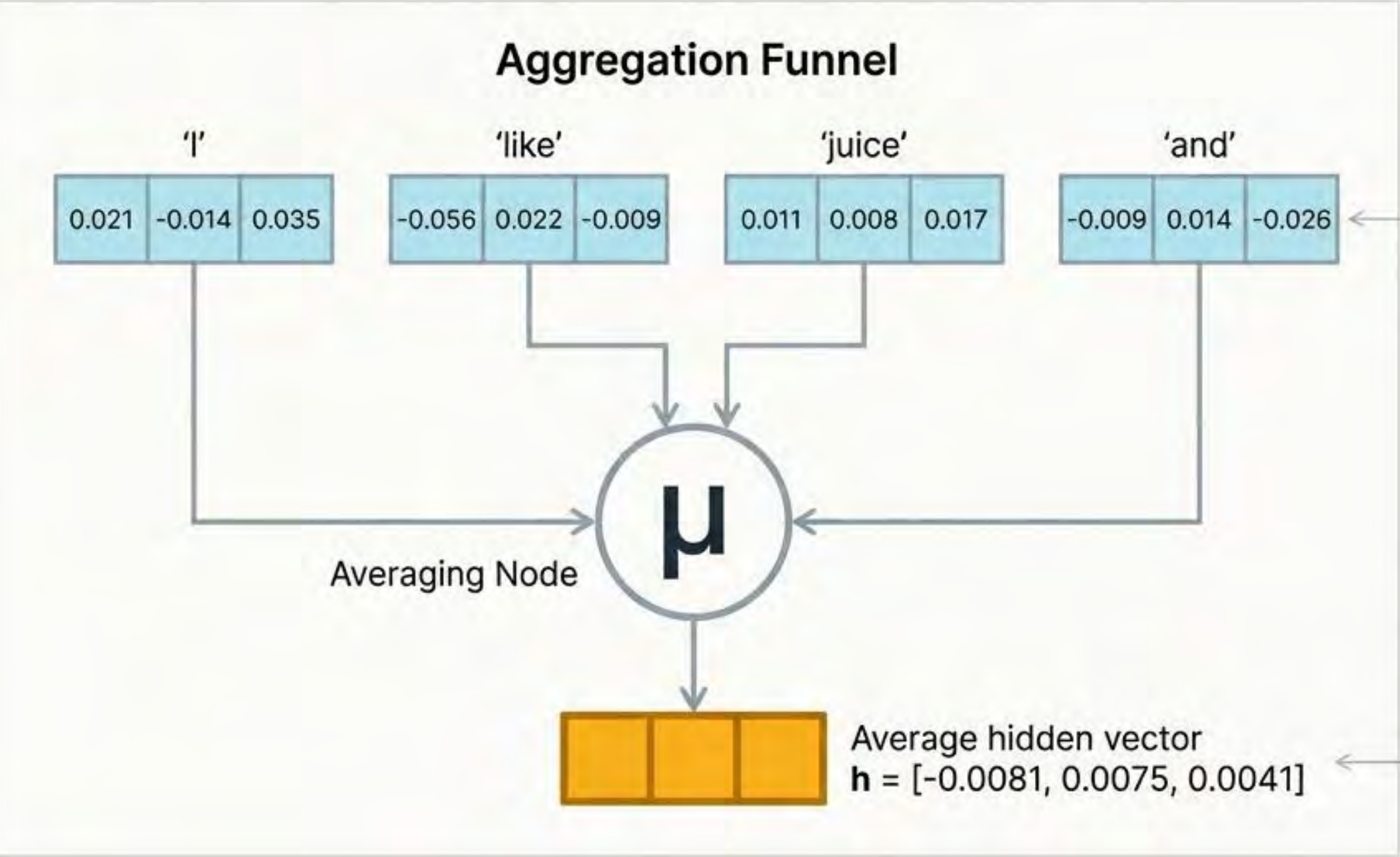


Multiplying by a one-hot array does not compute a complex transformation. Because the vector contains a single 1, it simply acts as a pointer, retrieving the exact row from the W matrix.



# Step 3: The Bag of Words

$$h = \text{mean}(\text{context\_embeddings}, \text{axis}=0)$$



By performing an **element-wise averaging** operation on the projected vectors, the exact structural order of the sentence is permanently lost.

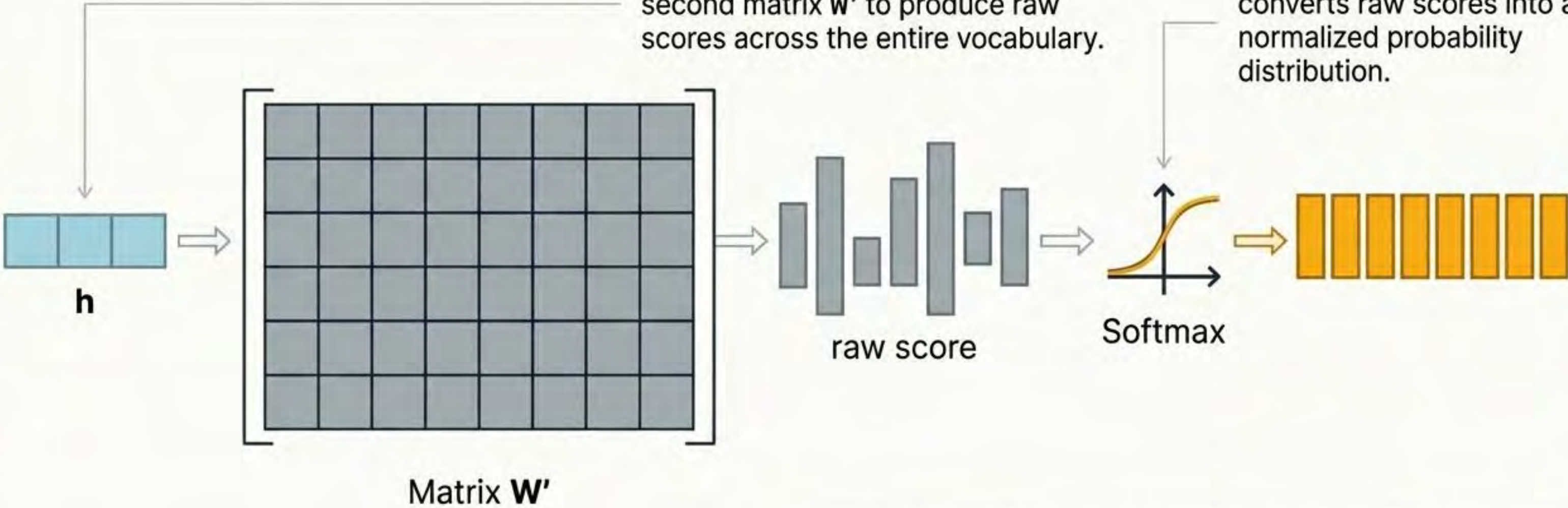
**Takeaway:** Distinct semantic signals of infrequent terms can be diluted by common neighbours.



# Step 4: Scoring and Prediction

The hidden vector  $\mathbf{h}$  is multiplied by a second matrix  $\mathbf{W}'$  to produce raw scores across the entire vocabulary.

A softmax activation converts raw scores into a normalized probability distribution.

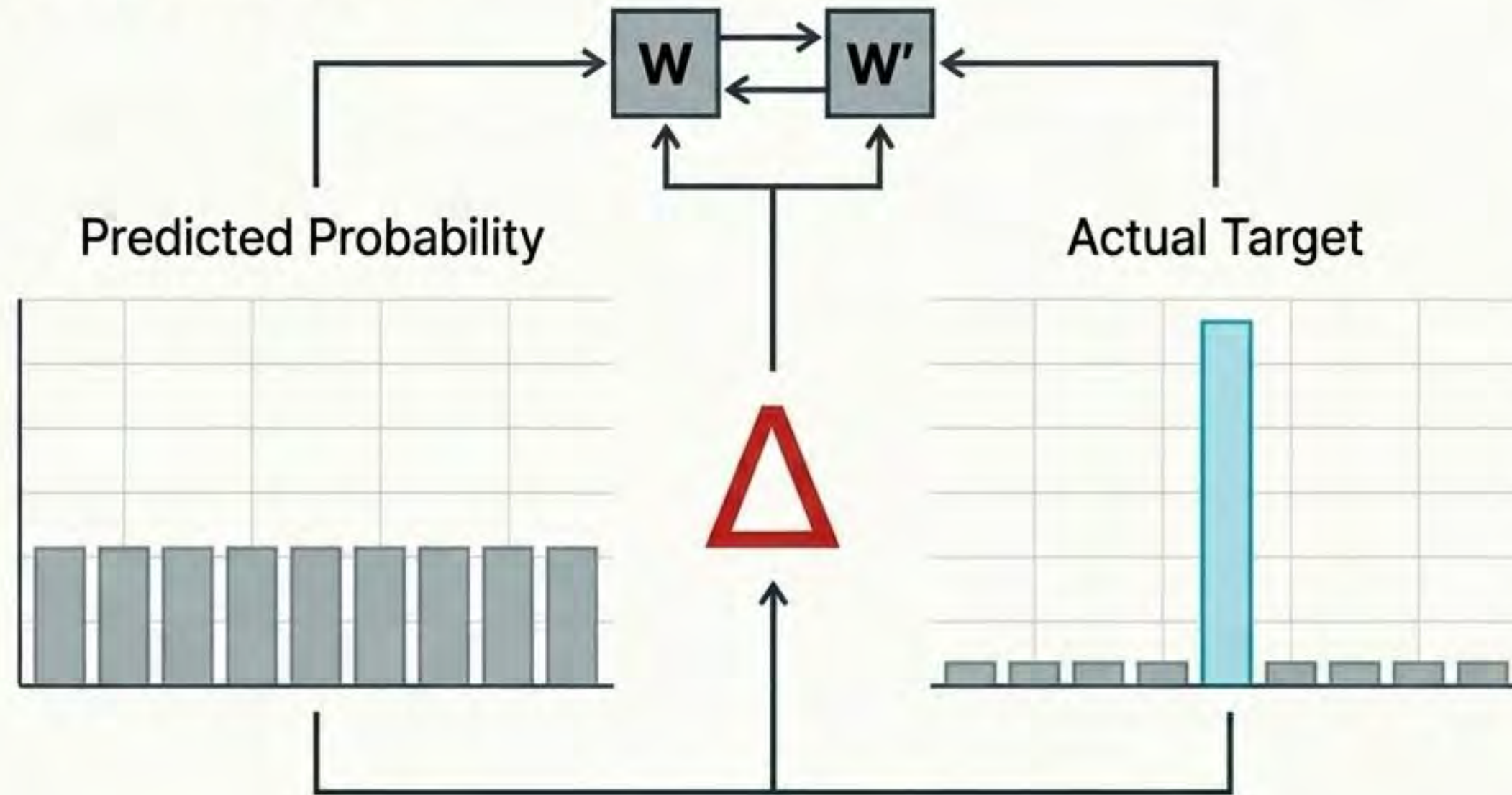


Pre-training, the prediction is effectively random. The model guesses orange with a **0.125016** probability.



# The Error Signal & Iterative Refinement

Data point:  
Cross-Entropy Loss  
 $= -\log(P(\text{target} | \text{context}))$   
 $= 2.079313.$

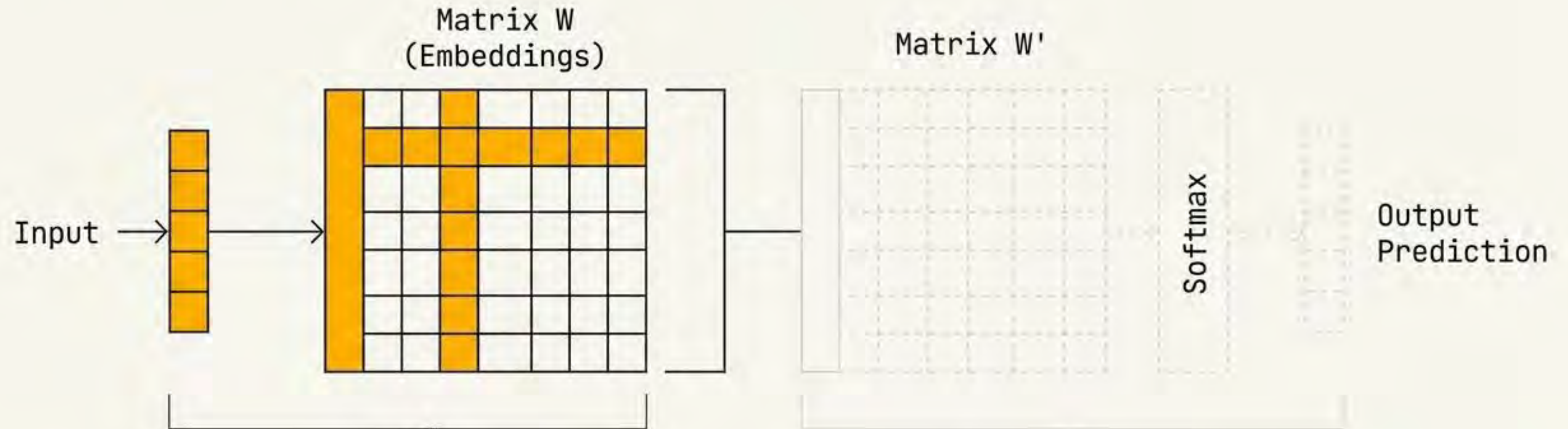


The model evaluates its random prediction using cross-entropy loss. Through backpropagation, gradients adjust the weight matrices in – a direction that reduces this discrepancy.

**Takeaway:** With each iteration, initially random vectors are transformed into meaningful geometric structures.



# The Prediction is a Proxy



## Insight

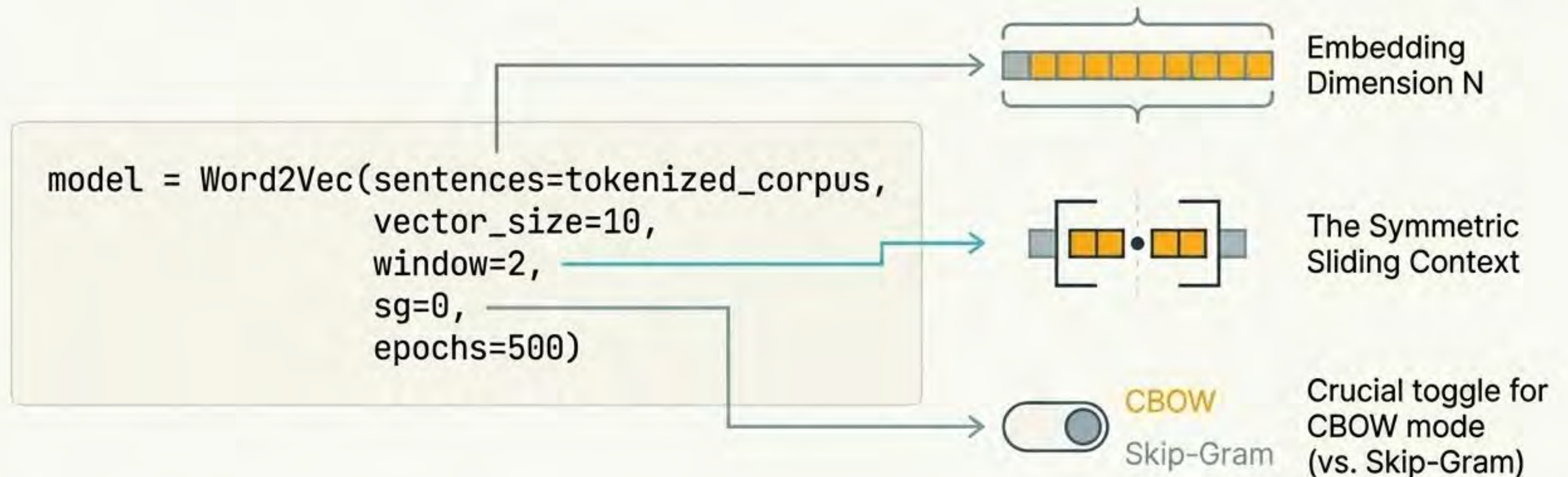
CBOW is a self-supervised system. The objective is not to predict the correct word in isolation. The prediction task acts solely as a proxy to force the network to learn geometric relationships.

## Takeaway

Once training completes, the output layer is discarded. The true product is the hidden layer weight matrix—the stored embeddings.



# Transitioning to Industry Practice

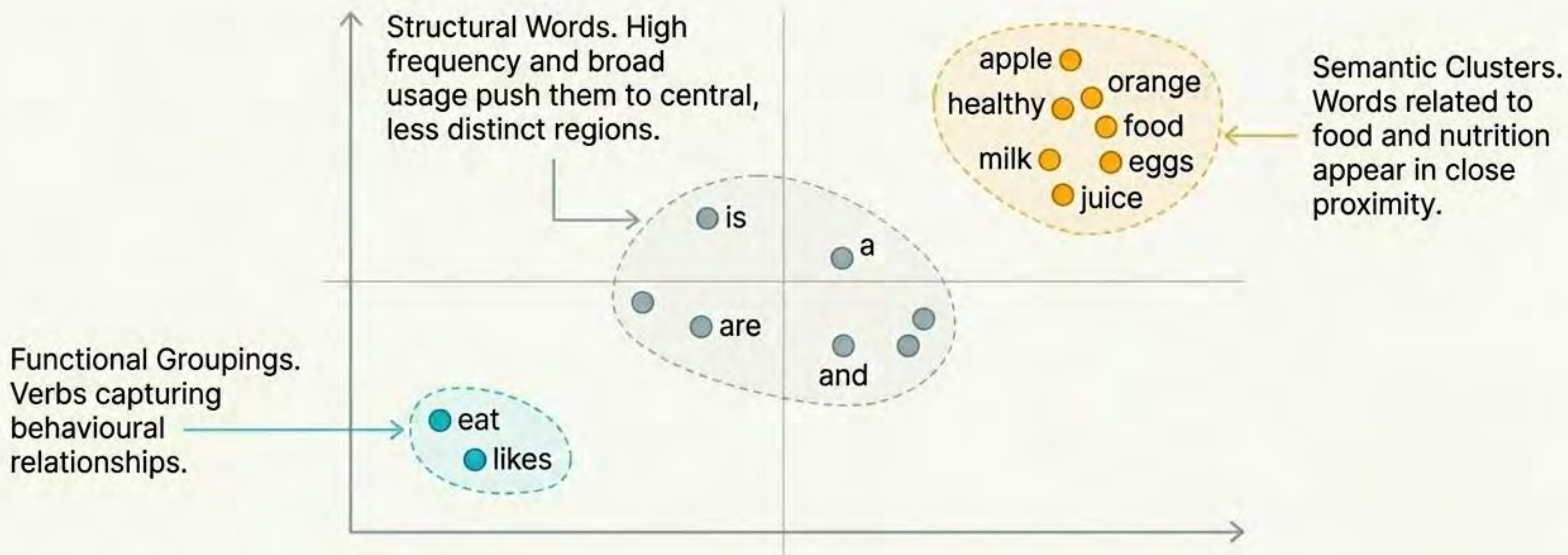


## Takeaway

Abstraction does not alter the underlying geometry. The library internally generates pairs, runs forward passes, and updates weights.



# The Geometric Matrix Visualised



## Takeaway

The model successfully learned context-driven semantic relationships from raw text.



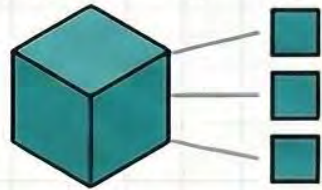
# CBOW: Conceptual Anchors and Representational Limits

The CBOW model represents a shift from symbolic language to learned geometric relationships. By predicting a target word from its surrounding context, the model transforms raw text into dense semantic embeddings.

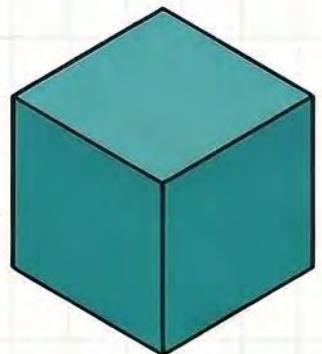
## THE LOGIC OF CONTEXT

### Context Window Size

**Small**  
(e.g., size 2)



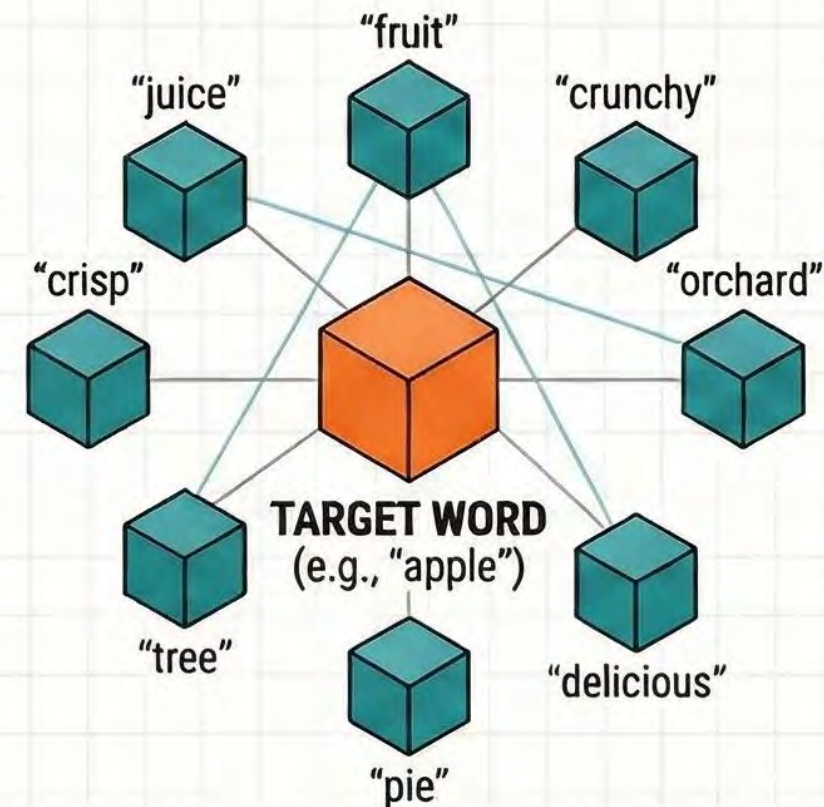
Syntactic structure and local dependencies.  
**Primary Focus:**  
Immediate Neighbours



**Large**  
(e.g., size 5+)

Semantic relationships and thematic clusters.  
**Primary Focus:**  
Broader Text Span

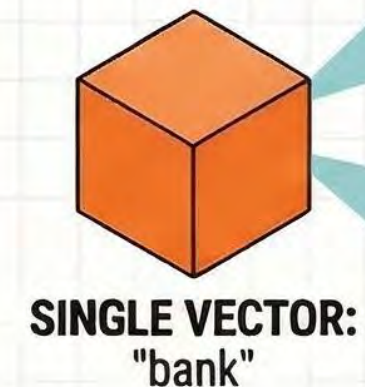
### Meaning is Learned from "Company"



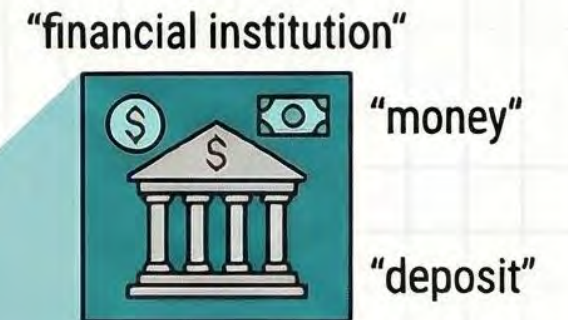
Meaning is not intrinsic to words but emerges from usage patterns and relationships.

## REPRESENTATIONAL CONSTRAINTS

### The Polysemy Barrier



#### Averaged Semantic Signals



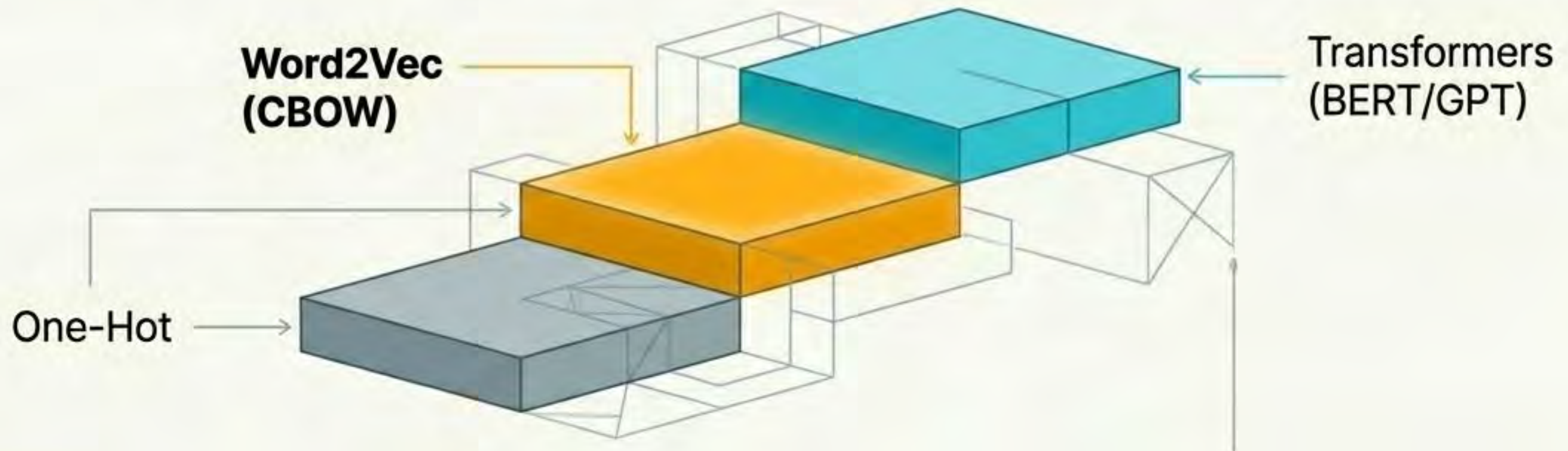
Example: A financial "bank" versus a riverbank. Single vectors cannot distinguish between multiple meanings.



Distinct meanings merge into one representation, leading to a permanent loss of contextual nuance.



# Geometry as the Language of Semantics



**Meaning from Usage:** Language is a system of relationships, not isolated definitions.

**Self-Supervised Foundation:** Raw text automatically provides its own training labels, a principle that underpins modern Large Language Models.

CBOW introduces a paradigm that extends far beyond Word2Vec: words are defined not by what they are, but by the company they keep.



# Support Independent Research: Fueling Discovery

## The Financial Barriers to Research



### Essential Infrastructure Costs

Funds are required for software licenses, platform access, and database subscriptions.



### Publication & Quality Assurance

Contributions cover article processing charges, professional editing tools, and private reviewers.



## How to Support the Work



### Discretionary Contributions

Support is entirely voluntary and based on the value found in the sessions.



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